Computable Functions

Def

A function $f : \Sigma_1^* \longrightarrow \Sigma_2^*$ is computable if there exists a Turing Machine s.t. for all $\omega \in \Sigma_1^*$, TM M on input ω halts with $f(\omega)$ on tape.

(q_{acc} & q_{rej} replaced with q_{halt})

f is a total function $\longrightarrow$ defined on all domain (Σ_1^*)

Could consider f as a partial function, i.e., f is undefined on some inputs
M doesn't halt if f is undefined.

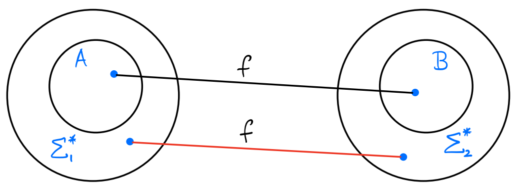
Mapping Reducibility

Consider $A \subseteq \Sigma_1^*$ & $B \subseteq \Sigma_2^*$

$A \leq_m B$ ↳ mapping reduction

$A \leq_T B$ ↳ Turing reduction

$A \leq_m B$ if there exists a computable $f : \Sigma_1^* \longrightarrow \Sigma_2^*$, such that
for all $\omega \in \Sigma_1^*$, $\omega \in A \iff f(\omega) \in B$



Yes instances of A map to Yes instances of B.
No instances of A map to No instances of B.

Properties

1. $A \leq_{m/T} B$ & B is decidable $\Rightarrow$ A is decidable
2. $A \leq_{m/T} B$ & A is undecidable $\Rightarrow$ B is undecidable

Recall Membership $= \{ \langle M, \omega \rangle \mid \langle M \rangle \text{ is a TM } \& \ M \text{ accepts } \omega \} = MP$

Is $MP \in$ Turing Recognizable? Yes!

Is $MP \in$ co-Turing Recognizable? No!

Note $\overline{MP} \leq_T MP$ & $MP \leq_T \overline{MP}$ Is $\overline{MP} \leq_m MP$? No! Why?

3. $A \leq_m B$ & B is Turing-Recognizable (TR) $\Rightarrow$ A is TR

AFSOC $\overline{MP} \leq_m MP \Rightarrow \overline{MP}$ is TR $\Rightarrow$ MP is co-TR $\Rightarrow$ MP is decidable
(Contradiction)

4. $A \le_m B$ & A is not TR $\Rightarrow$ B is not TR

5. $A \le_m B \Leftrightarrow \overline{A} \le_m \overline{B}$

6. $A \le_m B$ & $B \le_m C$ $\Rightarrow$ $A \le_m C$

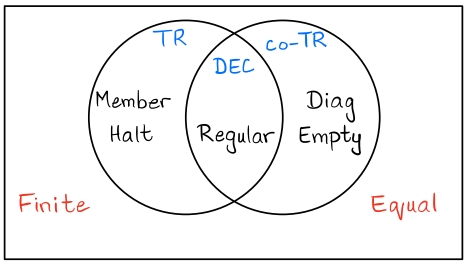
Additional examples of languages:

Diag $= \{ \langle M \rangle \mid \langle M \rangle \notin L(M)$ & M is a TM $\}$

Empty $= \{ \langle M \rangle \mid L(M) = \phi$ & M is a TM $\}$

Finite $= \{ \langle M \rangle \mid L(M)$ is finite & M is a TM $\}$

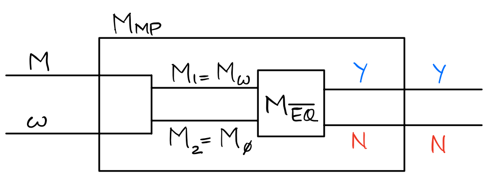
EQ = Equal $= \{ \langle M_1, M_2 \rangle \mid L(M_1) = L(M_2)$ & M_1 & M_2 are TMs $\}$



Equal is neither TR nor co-TR.

i) Equal $\notin$ TR

$\overline{MP} \le_m$ Equal $\Rightarrow$ MP $\le_m \overline{\text{Equal}}$



Recall for M_ω, $\omega \in L(M) \Leftrightarrow L(M_\omega) = \Sigma^*$
$\omega \notin L(M) \Leftrightarrow L(M_\omega) = \phi$

Make M_2, s.t., $L(M_2) = \phi$

$M_{\overline{EQ}}$ accepts $\Rightarrow L(M_\omega) \ne L(M_\phi)$
$\Rightarrow L(M_\omega) = \Sigma^*$
$\Rightarrow \omega \in L(M)$

$M_{\overline{EQ}}$ rejects $\Rightarrow L(M_\omega) = L(M_\phi)$
$\Rightarrow L(M_\omega) = \phi$
$\Rightarrow \omega \notin L(M)$

Similarly, prove ii) Equal $\notin$ co-TR

Gödel's Incompleteness Theorems

Developments in 19^{th} century concluded that all math proofs can be formally stated using First Order Logic (FOL) & a deductive calculus.

More precisely Gödel's Completeness Theorem states that starting from a set of axioms, you can prove anything that logically follows these axioms by applying the inference rules of FOL.

The Incompleteness Theorems allow us to distinguish between truth versus something provable from the axioms. They were a blow to Hilbert & others who wanted to formalize all of mathematics.

For example, Presburger arithmetic (1929) is a theory of natural numbers $\mathbb{N}$ with only addition over them.

Presburger Arithmetic – Theory $(\mathbb{N}, +)$

Have access to constants 0 & 1, and the binary function $\text{Plus}(\ ,\ )$.

Axioms of the Theory

1. $\forall x \ \neg(0 = \text{Plus}(x, 1))$
2. $\forall x \ \forall y \ (\text{Plus}(x, 1) = \text{Plus}(y, 1)) \Rightarrow x = y$
3. $\forall x \ \text{Plus}(x, 0) = x$
4. $\forall x \ \forall y \ (\text{Plus}(x, \text{Plus}(y, 1)) = \text{Plus}(\text{Plus}(x, y), 1)$

(Induction) 5. Let $S(x)$ be a formula in Presburger arithmetic with free variable x.
$S(0) \wedge \forall x (S(x) \Rightarrow S(x+1)) \Rightarrow \forall y \ S(y)$

Theorem

Theory $(\mathbb{N}, +)$ is decidable.

Think of this limited theory in a similar way as we have DFAs for computability.

In contrast, Theory $(\mathbb{N}, +, \times)$ is undecidable.
It is sufficiently expressive enough such that the Incompleteness Theorems apply.

Goals with the Axioms

1. Computable Axioms

$L = \{ \omega \mid \omega \text{ is an axiom} \}$ is decidable.

2. Consistency Let $A_1, A_2, \ldots, A_m$ be axioms.

The system is inconsistent if we can deduce both S & $\neg S$.

Can we axiomatize all math?

Frege proposed axioms of set theory.
Russell showed that these axioms allow us to define:

$D = \{ x \mid x \notin x \}$

Now, if $D \in D$ then $D \notin D$
if $D \notin D$ then $D \in D$ } Inconsistent

3. Soundness Let $A_1, A_2, \ldots, A_m$ be axioms.

If every statement S we can conclude is true, then system is sound.
Soundness $\Rightarrow$ Consistency

4. Completeness Let $A_1, A_2, \ldots, A_m$ be axioms.

If for all sentences S, either S or $\neg S$ is deducible from axioms then it is complete.

Completeness $\longrightarrow$ atleast one of S or $\neg S$

Consistency $\longrightarrow$ at most one of S or $\neg S$

Common to suffer from incompleteness when trying to axiomatize a branch of mathematics.

Zermelo-Fraenkel (ZF) set theory with axiom of choice (ZFC) is the standard axiomatic set theory today for the foundation of mathematics.

Interplay between Turing & Gödel

Recall $Halt = \{ \langle M, x \rangle \mid M \text{ halts on } x \}$ is undecidable.

Idea for M_H

```
for k = 1, 2, 3, ...
    for all strings P of length k
        if P is a valid ZFC + FOL proof of M halts on x
            return Yes
        if P is a valid ZFC + FOL proof of M loops
            return No
```

Since H is undecidable, there exists some M & x, such that ZFC + FOL cannot prove M halts on x OR M loops

But M_H must either halt or loop. One option has to be true.

So, we conclude that
There is a math statement that cannot be proved in ZFC + FOL. (ZFC is not complete)
OR
ZFC is not sound, i.e., proves a False statement, e.g., proves M halts when it loops.

Hence, ZFC cannot be both complete & sound.

Gödel's First Incompleteness Theorem

Any axiomatic system that is sufficiently expressive (i.e., can define TMs, e.g., ZFC) & has computable axioms cannot be both complete & sound.

Provability versus Truth

How can we say a statement S is True if we cannot prove it.

Let's add it as an axiom.

The modified system will have a different True statement that is not provable.

Add all True statements as axioms.

The set of axioms is uncomputable.

Theorem

Any axiomatic system that is sufficiently expressive (i.e., can define TMs, e.g., ZFC) & has computable axioms cannot be both complete & consistent.

Lemma

If a TM M has a t-step execution, then there is a proof of it in ZFC.

In particular, if $M(x)$ halts, then there is a proof of $M(x)$ halts.

Proof is the trace of computation history.

Proof of Theorem

Assume ZFC is consistent.

Let D be a TM on input ⟨M⟩:

```
for k = 1, 2, 3, ...
    for all strings P of length k
        if P is a valid ZFC + FOL proof of M(<M>) halts
            go right forever
        if P is a valid ZFC + FOL proof of M(<M>) loops
            return
```

What can ZFC prove about D(⟨D⟩)?

By consistency assumption either D(⟨D⟩) halts
OR D(⟨D⟩) loops

Perhaps ZFC can prove D(⟨D⟩) loops.

Then D on input ⟨D⟩ will find proof and halt.

But if D(⟨D⟩) halts, then by lemma above ZFC can prove D(⟨D⟩) halts

Violate consistency

Perhaps ZFC can prove D(⟨D⟩) halts
Then D on input ⟨D⟩ will run some t-steps, find proof & then loop
Thus, D(⟨D⟩) loops

Violate consistency

So, ZFC is incomplete.

In truth, does D(⟨D⟩) halt or does D(⟨D⟩) loop?

Actually, D(⟨D⟩) loops! Why? Since ZFC is incomplete, D cannot prove either statement & hence loops over the infinite strings.

Doesn't this give a proof of D(⟨D⟩) loops?

No! We get a proof of 'ZFC is consistent ⟹ D(⟨D⟩) loops'

To avoid contradiction, we must conclude that

ZFC cannot prove that 'ZFC is consistent'.

Gödel's Second Incompleteness Theorem

Assume ZFC is consistent. Then, not only is it incomplete, here is a True statement it cannot prove, i.e., ZFC is consistent.